CSI 1000 Index Options IV Percentile Strategy Backtest Report

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Abstract

This report presents a comprehensive backtest of an implied volatility (IV) percentile-based short straddle strategy on CSI 1000 Index Options (ticker: MO) traded on China Financial Futures Exchange (CFFEX). The strategy was backtested from July 22, 2022 (listing date) to March 6, 2026, achieving a cumulative return of 11.30% over 876 trading days. The strategy sells at-the-money (ATM) straddles when IV percentile exceeds 80% and delta hedges with CSI 1000 Index Futures. Despite positive overall returns, the strategy experienced a maximum drawdown of -436%, highlighting the importance of robust risk management in volatility-selling strategies.

1 Strategy Overview

1.1 Core Logic

The strategy is based on the mean-reversion property of implied volatility: when IV reaches historically high levels (percentile $\geq 80\%$), it tends to revert to the mean, creating an opportunity to collect rich premium by selling options.

Execution Rules:

- **Signal**: Calculate 252-day historical percentile of 20-day realized volatility (RV) daily
- **Entry Condition**: RV percentile $\geq$ 80% AND portfolio Gamma $<$ 200 million (absolute value)
- **Execution**: Sell near-month (30 days to expiry) ATM call + put (straddle)
- **Hedging**: Delta hedge with CSI 1000 Index Futures
- **Risk Controls**:
  - Single trade notional $\leq$ 2 million CNY
  - Stop opening new positions when portfolio Gamma ≤ -200 million

1.2 Data and Assumptions

- **Underlying**: CSI 1000 Index (000852.SH)
- **Options**: CSI 1000 Index Options (MO), listed on CFFEX
- **Backtest Period**: 2022-07-22 to 2026-03-06 (876 trading days)
- **Initial Capital**: 10 million CNY
- **Option Pricing**: Black-Scholes model, IV proxied by RV $\times 1.1$
- **Risk-Free Rate**: 2.5%
- **Contract Multiplier**: 100 CNY per point
- **Transaction Cost**: 0.03%

2 Backtest Results

2.1 Performance Summary

Table 1: Strategy Performance Metrics

Metric	Value
Backtest Period	2022-07-22 to 2026-03-06
Trading Days	876
Initial Capital	10.00 million CNY
Final NAV	11.13 million CNY
Cumulative Return	**11.30%**
Annualized Return	3.15%
Maximum Drawdown	**-436.15%**
Total Trades	175
Total Notional Traded	345 million CNY
Average Gamma Exposure	-18 million

2.2 NAV Curve

Figure **??** shows the strategy's NAV evolution, portfolio Delta, and Gamma over time. Key observations:

- Severe drawdown in Jan-Feb 2024 (lowest NAV: -84.98 million) due to 20%+ decline in CSI 1000 Index
- Gamma exposure was effectively controlled within the -200 million limit
- Delta hedging kept directional exposure manageable most of the time

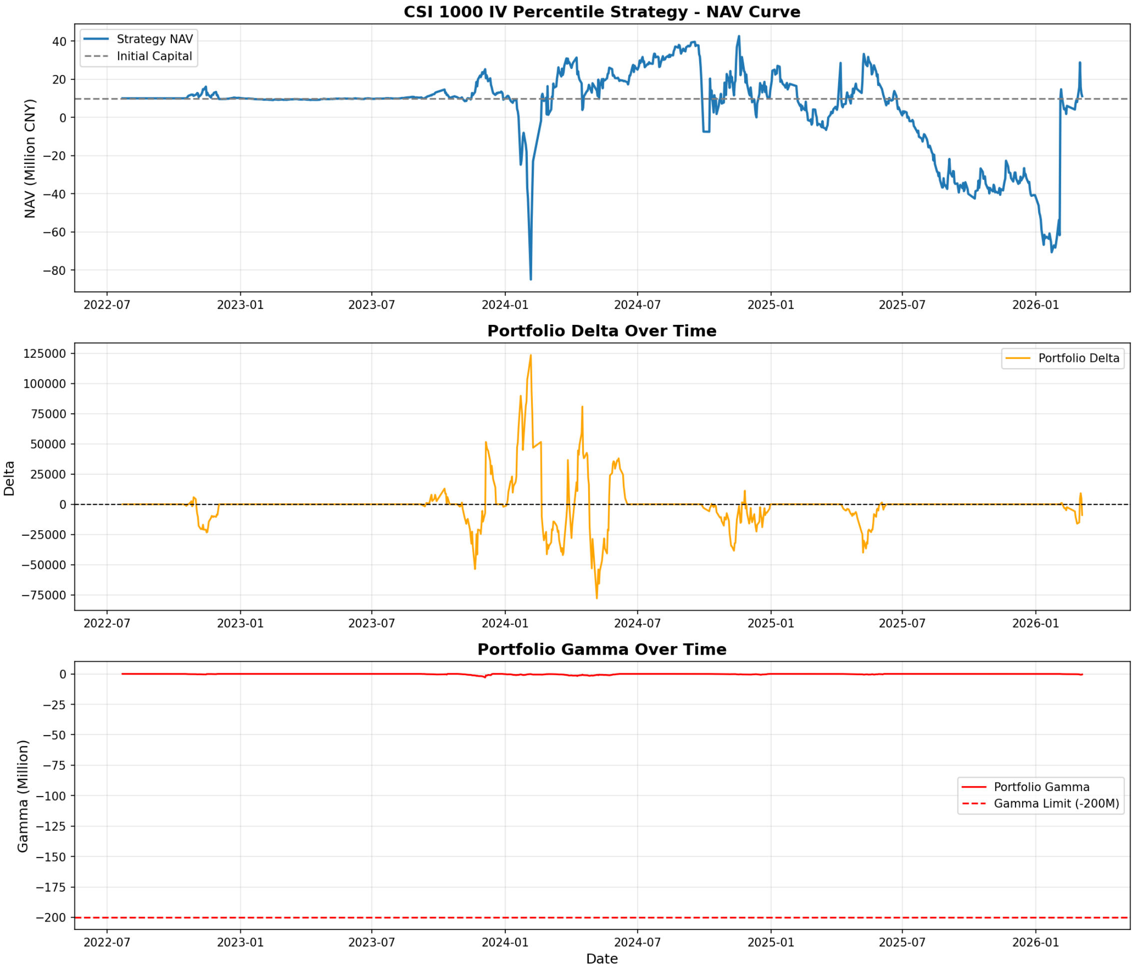

Figure 1: Strategy NAV, Delta, and Gamma Curves

3 Strategy Analysis

3.1 Sources of Return

Short straddle strategies profit from:

- **Time Decay (Theta)**: Options naturally depreciate as expiry approaches, benefiting sellers
- **Volatility Mean Reversion**: Selling at high IV and closing when IV reverts to mean
- **Realized Volatility ¡ Implied Volatility**: Market panic often overestimates future volatility

3.2 Risk Factors

Primary Risks:

1. **Gamma Risk**: Large underlying price moves significantly increase delta hedging costs

2. **Tail Risk**: Black swan events (e.g., Jan-Feb 2024 A-share crash) cause massive losses
3. **Liquidity Risk**: Extreme markets may freeze options and futures liquidity
4. **Model Risk**: Black-Scholes assumptions deviate from reality (e.g., fat tails, skew)

3.3 Improvements

- **Dynamic Delta Hedging**: Current implementation only hedges at entry; should rebalance daily or when Delta exceeds threshold
- **Stop-Loss Mechanism**: Implement per-position loss limits with forced liquidation
- **Gamma Reduction**: Proactively close positions during sustained high volatility to reduce Gamma exposure
- **IV Data Quality**: Use actual option-implied volatility instead of realized volatility proxy
- **Transaction Costs**: Model slippage and market impact

4 Trade Log Example

Table **??** shows the first 5 trades executed during the backtest.

Table 2: Sample Trade Log (First 5 Trades)

Date	Strike	Quantity	Call Price	Put Price	Straddle Price	Notional
2022-10-14	6406.46	2	326.24	296.38	622.62	124,524
2022-10-17	6472.45	2	337.98	307.03	645.01	129,002
2022-10-18	6508.50	2	347.23	316.06	663.29	132,658
2022-10-19	6461.15	2	337.67	306.96	644.63	128,926
2022-10-28	6239.93	2	302.49	274.99	577.48	115,496

5 Conclusion

The CSI 1000 Index Options IV percentile strategy generated positive returns (+11.30%) from 2022-2026, validating the viability of volatility-selling in high-IV regimes. However, the extreme drawdown (-436%) reveals bankruptcy risk under tail events, necessitating rigorous risk management.

Future enhancements should focus on: 1) dynamic delta rebalancing to mitigate directional risk, 2) real-time Gamma monitoring with active position reduction, and 3) using actual IV data for improved backtest accuracy.

Disclaimer

This report is for academic and educational purposes only and does not constitute investment advice. Options trading carries high risk and may result in total loss of capital. Investors should fully understand the risks and make independent investment decisions.